CCC-C-426c
January 3, 1966
SUPERSEDING
Fed. Spec. CCC-C-426b
December 13, 1962

FEDERAL SPECIFICATION

# CLOTH, DRILL, COTTON

*This specification was approved by the Commissioner, Federal Supply Service, General Services Administration, for the use of all Federal agencies.*

## 1. SCOPE AND CLASSIFICATION

**1.1 Scope.** This specification covers cloth, drill, cotton.

**1.2 Classification.**

**1.2.1** *Types and classes.* The cloth shall be of the following types and classes (see 6.2):

Type I—30 inches[1] 72 by 60, 2.50 yd. (commercial designation).
Type II—30 inches[1] 72 by 48, 2.50 yd. (commercial designation).
Type III—30 inches[1] 72 by 48, 2.85 yd. (commercial designation).

Class 1—Greige.
Class 2—Desized and preshrunk.
Class 3—Bleached white or dyed and preshrunk.

[1]Prorata widths of like construction are acceptable when specified (see 6.2).

## 2. APPLICABLE DOCUMENTS

**2.1 Specifications and standards.** The following specifications and standards, of the issues in effect on date of invitation for bids or request for proposal, form a part of this specification to the extent specified herein:

*Federal Specifications*:

V-T-276—Thread, Cotton.
CCC-T-191—Textile Test Methods.
PPP-P-51—Packaging, Packing, and Marking of Textile Fabrics (Woolens, Worsteds, Cottons, Silks, and Synthetics).

*Federal Standards*:

Fed. Std. No. 4—Glossary of Fabric Imperfections.
Fed. Std. No. 102—Preservation, Packaging, and Packing Levels.

(Activities outside the Federal Government may obtain copies of Federal Specifications, Standards, and Handbooks as outlined under General Information in the Index of Federal Specifications and Standards and at the prices indicated in the Index. The Index, which includes cumulative monthly supplements as issued, is for sale on a subscription basis by the Superintendent of Documents, U.S. Government Printing Office, Washington, D.C., 20402.

(Single copies of this specification and other product specifications required by activities outside the Federal Government for bidding purposes are available without charge at the General Services Administration Regional Offices in Boston, New York, Washington, D.C., Atlanta, Chicago, Kansas City, Mo., Dallas, Denver, San Francisco, Los Angeles, and Seattle, Wash.

(Federal Government activities may obtain copies of Federal Specifications, Standards, and Handbooks and the Index of Federal Specifications and Standards from established distribution points in their agencies.)

*Military Standard*:

MIL-STD-105—Sampling Procedures and Tables for Inspection by Attributes.

(Copies of Military Specifications and Standards required by contractors in connection with specific procurement functions should be obtained from the procuring activity or as directed by the contracting officer.)

**2.2 Other publications.** The following document forms a part of this specification to



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the extent specified herein. Unless otherwise indicated, the issue in effect on date of invitation for bids or request for proposal shall apply.

Rules and Regulations Under the Textile Fiber Products Identification Act.

(Copies may be obtained without charge from the Federal Trade Commission, Washington, D.C., 20580).

## 3. REQUIREMENTS

**3.1 Standard sample.** When a standard sample is available, the cloth shall match the standard sample for shade and shall be equal to or better than the standard sample with respect to all characteristics for which the standard sample is referenced (see 6.3).

**3.2 Material.**

**3.2.1** *Yarn.* The yarn shall be made from clean cotton, carded, drawn, and spun into single yarn for both the warp and filling.

**3.3 Color.** Classes 1 and 2 cloths shall be natural color. Class 3 cloth shall be bleached or dyed as specified (see 6.2). The use of sulfur dyes or dyes containing elementary sulfur or compounds capable of oxidation to sulfuric acid is prohibited. The dyestuffs shall be chosen and applied so that the dyed cloth shall show no more free or sulfide sulfur than the standard sample when tested as specified in 4.4. When no standard sample is available, there shall be a minimum amount of staining of the test paper when tested as specified in 4.4.

**3.3.1** *Matching.* The dyed cloth shall match the standard sample under natural (north sky) daylight or artificial daylight having a color temperature of 7500° Kelvin and shall be a good approximation to the standard sample under incandescent lamplight at 2800° Kelvin.

**3.3.2** *Colorfastness.* The dyed and finished cloth shall show colorfastness to laundering, wet-drycleaning, perspiration, crocking, and light equal to or better than the standard sample. When no standard sample is established or when the standard sample is not a reference for colorfastness, the dyed and finished cloth shall show good colorfastness to laundering, wet-drycleaning, perspiration, light, and no more crocking than Munsell neutral value 8.5 when tested as specified in 4.4.

**3.4 Physical requirements.** The finished cloth shall conform to the minimum requirements specified in table I when tested as specified in 4.4.

**3.4.1** *Width.* The width shall be as specified (see 6.2), and shall be the minimum acceptable width, inclusive of selvages.

**3.5 Nonfibrous materials.** The starch and protein content including chloroform-soluble and water-soluble material of the finished cloth shall not exceed the percentage for

TABLE I. *Physical requirements*

| Type | Class | Weight per square yard | Yarns per inch | | Breaking strength | | Weave (for all types) |
|---|---|---|---|---|---|---|---|
| | | | Warp | Filling | Warp | Filling | |
| | | *Ounces* | | | *Pounds* | *Pounds* | |
| I | 1 | 7.7 | 72 | 60 | 130 | 90 | 3 harness |
| | 2 | 7.5 | 74 | 58 | 120 | 85 | 2 left twill |
| | 3 | 6.5 | 74 | 56 | 105 | 70 | 1 |
| II | 1 | 7.7 | 72 | 48 | 130 | 85 | |
| | 2 | 7.5 | 74 | 46 | 120 | 72 | |
| | 3 | 6.5 | 74 | 44 | 105 | 62 | |
| III | 1 | 6.7 | 72 | 48 | 110 | 65 | |
| | 2 | 6.5 | 74 | 46 | 100 | 60 | |
| | 3 | 6.0 | 74 | 44 | 90 | 55 | |



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the applicable class as shown below when tested as specified in 4.4.

| Class | Percent |
|---|---|
| 1 | 12.0 |
| 2 | 5.0 |
| 3 | 2.0 |

**3.6 Shrinkage.** Classes 2 and 3 cloths shall be preshrunk to a residual shrinkage of not more than 2.0 percent in the direction of the warp and not more than 2.0 percent in the direction of the filling when tested as specified in 4.4. The preshrinking process used shall not be identified by name or trademark, either on the cloth, ticket, or package.

**3.7 Seam efficiency.** The cloth for all types and classes shall have a seam efficiency of not less than 80 percent when tested as specified (see 4.4).

**3.8 Length and put-up.** Unless otherwise specified (see 6.2), the cloth shall be in continuous lengths, each not less than 40 yards. Each length shall be put up in flat folded pieces as specified in PPP-P-51.

**3.9 Fiber identification.** Each piece shall be labeled, ticketed, or invoiced for fiber content in accordance with the Rules and Regulations Under the Textile Fiber Products Identification Act.

**3.10 Workmanship.** The finished cloth shall conform to the quality established by this specification. The demerit points per 100 square yards when calculated as specified in section 4 shall not exceed the established maximum point value.

## 4. QUALITY ASSURANCE PROVISIONS

**4.1 Responsibility for inspection.** Unless otherwise specified in the contract or purchase order, the supplier is responsible for the performance of all inspection requirements as specified herein. Except as otherwise specified, the supplier may utilize his own facilities or any commercial laboratory acceptable to the Government. The Government reserves the right to perform any of the inspections set forth in the specification where such inspections are deemed necessary to assure that supplies and services conform to prescribed requirements.

**4.2 Examination of the end item.** Examination of the end item shall be in accordance with the provisions of MIL-STD-105 and 4.2.2 through 4.2.5.

**4.2.1** *Component and material inspection.* In accordance with 4.1, components and materials shall be tested in accordance with all the requirements of referenced specifications, drawings, and standards unless otherwise excluded, amended, modified, or qualified in this specification or applicable procurement documents. In addition, the supplier shall furnish a certificate of compliance for singles cotton yarn.

**4.2.2** *Yard-by-yard examination.* Thirty yards from each piece in the sample shall be examined on the face side. The sample shall consist of 25 pieces taken from 25 containers. All defects as defined in section I of Fed. Std. No. 4, which are clearly noticeable at normal inspection distance (3 feet), shall be scored and assigned demerit points as listed in 4.2.2.1, except that only those knots and slubs which exceed the limits shown on figure 1 shall be scored. No folded linear yard shall be penalized more than 4 points. The lot shall be unacceptable if the points per hundred square yards exceeds the following values:

Class 1—45.00 points.
Class 2—45.00 points.
Class 3—35.00 points for dyed cloth.
42.00 points for bleached cloth.

Point computation shall be as follows:

$$\frac{\text{Total points scored in sample size} \times 3{,}600}{\text{Contracted width of cloth (inches)} \times 750} = \text{Points per 100 square yards}$$

**4.2.2.1** *Demerit points.* Demerit points shall be assigned as follows:

For defects 3 inches or less in any dimension — one point.



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For defects exceeding 3 inches, but not exceeding 6 inches in any dimension — two points.

For defects exceeding 6 inches, but not exceeding 9 inches in any dimension — three points.

For defects exceeding 9 inches in any dimension — four points.

Note: The following defects, when present, shall be scored four points for each yard in which they occur:

Baggy, ridgy, or wavy cloth.

Objectionable odor.

Width less than specified.

Poor dye penetration, mottled, streaky, or cloudy.

4.2.3 *Examination for length.*

4.2.3.1 *Individual pieces.* During the yard-by-yard examination, each piece in the sample shall be examined for length. Any length found to be less than the minimum specified or more than two yards less than the length marked on the ticket shall be considered a defect with respect to length. The lot shall be unacceptable if two or more pieces in the sample are defective in respect to length.

4.2.3.2 *Total yardage in sample.* The lot shall be unacceptable if the total of the actual lengths of pieces in the sample is less than the total of the lengths marked on the ticket.

4.2.4 *Examination for shade.* During the yard-by-yard examination, each piece in the sample shall be examined for shade. Any piece in the sample off shade, shaded side to side, side to center, or end to end shall be cause for rejection of the entire lot represented by the sample.

4.2.5 *Examination for identification of preshrinkage process and noncompliance with the Textile Fiber Products Identification Act.* During the yard-by-yard examination, each piece in the sample shall be examined for these defects. The lot shall be unacceptable if two or more pieces in the sample contain identification of the preshrinkage process by name or trademark on the cloth or ticket or are not labeled or ticketed in accordance with the Textile Fiber Products Identification Act.

**4.3 Examination of preparation for delivery requirements.** An examination shall be made in accordance with the provisions of PPP-P-51 to determine that packaging, packing, and marking requirements of section 5 are compplied with.

**4.4 Testing of the end item.** The methods of testing specified in CCC-T-191, wherever applicable, and as listed in table II shall be followed. The physical and chemical values specified in section 3 apply to the average of the determinations made on a sample unit for test purposes as specified in the applicable test method. The sample unit shall be 2-1/2 continuous yards, full width, of the class 1 cloth, and 4-1/2 yards, full width, of classes 2 and 3 cloths. The lot shall be unacceptable if one or more sample units fail to meet any test requirements specified. The lot size shall be expressed in units of 1 yard. All test reports shall contain the individual values utilized in expressing the final results. The sample size (number of sample units) shall be in accordance with the following:

| Lot size (yards) | Sample size |
|---|---|
| 800 or less | 2 |
| 801 up to and including 22,000 | 3 |
| 22,001 and up | 5 |

TABLE II. *Tests*

| Characteristic | Requirement reference | Test method |
|---|---|---|
| Sulfur content[1] | 3.3 | 4.4.1 |
| Colorfastness to: | | |
| Light | 3.3.2 | 5660 |
| Laundering | 3.3.2 | 5610 |
| Wet-drycleaning | 3.3.2 | 5622 |
| Crocking | 3.3.2 | 5651 |
| Perspiration | 3.3.2 | 5680 |
| Weight | 3.4 | 5041 |
| Yarns per inch | 3.4 | 5050 |
| Weave[1] | 3.4 | Visual |
| Breaking strength | 3.4 | 5100 |
| Nonfibrous material content | 3.5 | 2611 |
| Shrinkage in laundering | 3.6 | 5550 |
| Seam efficiency[2] | 3.7 | 5110 |

[1]One determination shall be made from each sample



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unit and the result shall be reported as "pass" or "fail".

[2] The thread shall conform to type IA1 of V-T-276. The needle size shall be 0.049 ± 0.001-inch, and the top (needle) thread shall be 24/3, and the bottom (looper) thread shall be 60/3.

4.4.1 *Test for sulfur content.*

4.4.1.1 *Solution (acid stannous chloride).* Dissolve 100 grams (g.) of stannous chloride crystals in 100 ml. of hydrochloric acid (35 percent concentration) and add 50 ml. of distilled water.

4.4.1.2 *Procedure.* One to 1.5 g. of the finished cloth, cut into small pieces, shall be used for the test. Place the sample in a test tube of suitable size and cover with the stannous chloride solution. Place a filter paper which has been moistened with one drop of lead-acetate solution (made by dissolving 5 g. of reagent-grade lead acetate to 100 ml. of distilled water; add a drop of glacial acetic acid to clear up if solution is cloudy) over the mouth of the tube (punch needle hole in paper). Heat the test tube slowly over a low flame until the solution is boiling. Solution shall not be boiled for more than 10 to 15 seconds, as prolonged boiling will reduce the sensitivity of the test. A brown to black stain appearing on the paper indicates the presence of sulfur.

4.4.1.3 *Evaluation.* The stained filter paper shall then be compared with that obtained from the standard. If the stained area is equal to or lighter than the standard in the degree of staining, the sample shall be considered satisfactory. If the degree of staining by the sample is in excess of the standard, the sample shall be considered unsatisfactory. When no standard sample is available there shall be a minimum of staining on the paper.

## 5. PREPARATION FOR DELIVERY

Fed. Std. No. 102 shall be referred to for definitions and applications of the various levels of packaging and packing for civil agency procurement.

**5.1 Packaging.** Packaging shall be level A or C (see 6.2 and 6.2.1).

5.1.1 *Levels A and C.* The cloth, put up as specified, shall be packaged in accordance with the applicable requirements for class 3 fabrics of PPP-P-51.

**5.2 Packing.** Packing shall be level A, B, or C (see 6.2 and 6.2.1).

5.2.1 *Levels A, B, and C.* The cloth shall be packed in accordance with the applicable requirements of PPP-P-51.

**5.3 Marking.** In addition to any special marking required by the contract or order, shipments shall be marked in accordance with the requirements of PPP-P-51.

## 6. NOTES

**6.1 Intended use.** The cloth covered by this specification is primarily intended for use in clothing and equipage items.

**6.2 Ordering data.** Purchasers should exercise any desired options offered herein, and procurement documents should specify the following:

(*a*) Title, number, and date of this specification.
(*b*) Type and class required (see 1.2.1).
(*c*) Color of cloth required (see 3.3).
(*d*) Width of cloth required (see 3.4.1).
(*e*) Minimum acceptable length, if other than specified (see 3.8).
(*f*) Selection of applicable levels of packaging and packing (see 5.1 and 5.2).

6.2.1 In the preparation of contracts or orders, it must be noted that various options, choices, and alternatives, as indicated in PPP-P-51, may be exercised in the preparation for delivery of this cloth.

**6.3 Standard sample.** For access to standard sample, address the procuring office issuing the invitation for bids.

**6.4 Transportation description.** The transportation description and minimum weights applicable to the commodity are:



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*Rail*:

Cotton piece goods, finished, not otherwise indexed by name.
Carload minimum weight—any quantity.

*Motor*:

Cotton piece goods, not otherwise indexed, in the original piece.
Truckload minimum weight—any quantity, subject to Rule 115, National Motor Freight Classification.

**CUSTODIANS:**
**Army—GL**
**Navy—SA**
**Air Force—11**

*Review activities*:
**Army—MD**
**Navy—SA, MC**
**Air Force—11, 67**

*User activities*:
**Navy—YD, SH**

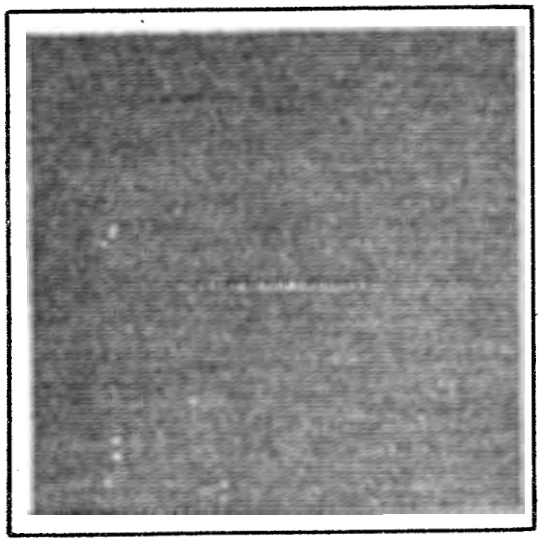
*Preparing activity*:
**Army—GL**

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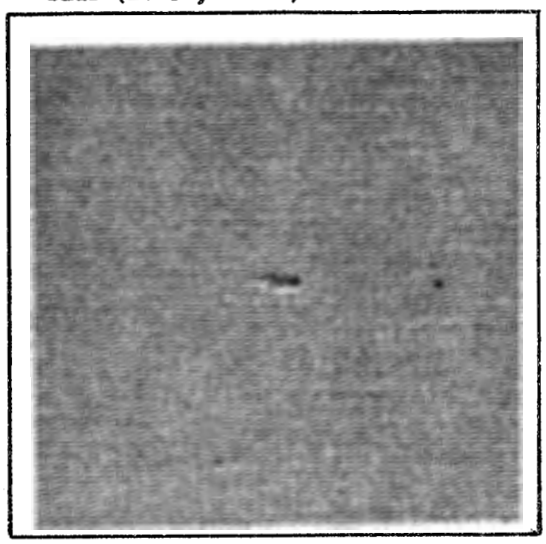


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Slub



Slub (Loose, Thick)



Lump or Knot (Fuzz)

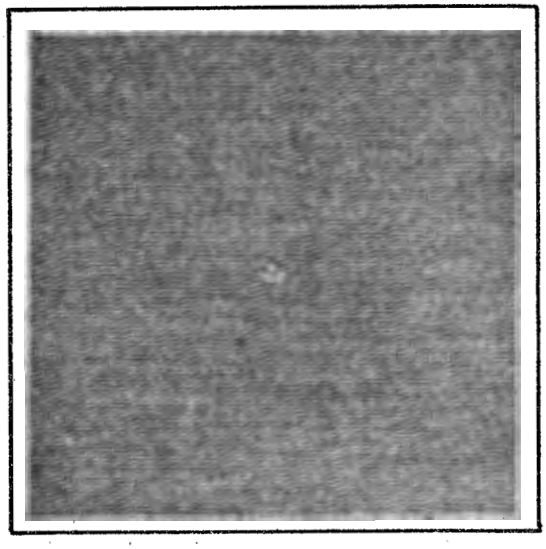


FIGURE 1. Limits of acceptability.

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